TI with Half Flux Quantum Revisited

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Abstract

The wormhole effect has been reported to stands as a bulk property of 3D TI, while this stands only when the half magnetic flux crosses through a single unit cell. When the flux is dispersive, the wormhole state generally contains non-zero energy.

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I. Basics

We begin with clarifying operators in polar coordinates. The coordinate transformation reads

$$\begin{cases} r = \sqrt{x^2+y^2} \\ \theta = \arctan(y/x) \end{cases}, \quad \begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}, \tag{1}$$

which leads to

$$\frac{\partial}{\partial x} = \frac{\partial r}{\partial x}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x}\frac{\partial}{\partial \theta} = \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta}, \tag{2a}$$

$$\frac{\partial}{\partial y} = \cdots \qquad = \sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial \theta}, \tag{2b}$$

and

$$\frac{\partial}{\partial x} + i\frac{\partial}{\partial y} = e^{i\theta}\left(\frac{\partial}{\partial r} + \frac{i}{r}\frac{\partial}{\partial \theta}\right), \tag{3a}$$

$$\frac{\partial}{\partial x} - i\frac{\partial}{\partial y} = e^{-i\theta}\left(\frac{\partial}{\partial r} - \frac{i}{r}\frac{\partial}{\partial \theta}\right). \tag{3b}$$

We can firstly look for the (canonical) angular momentum L_z in polar coordinates (we use $r_x = x,\ r_y = y$ and so on), which is defined base on the Cartesian coordinate $\boldsymbol{r}$ and its canonical momentum $\boldsymbol{p} = -i\hbar\nabla$ (basis ignored) as

$$\begin{aligned} L_z &= \epsilon_{3mn} r_m p_n = r_x p_y - r_y p_x \\ &= -i\hbar \left[r\cos\theta \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r} \frac{\partial}{\partial \theta} \right) - r\sin\theta \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta} \right) \right] \\ &= -i\hbar \partial_\theta. \end{aligned} \tag{4}$$

For the non-relativistic free particle, its Hamiltonian is proportional to

$$\begin{aligned} \frac{p^2}{-\hbar^2} &= \partial_i \partial_i \\ &= \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta} \right) \\ &+ \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r} \frac{\partial}{\partial \theta} \right) \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r} \frac{\partial}{\partial \theta} \right) \\ &= \cos^2\theta \partial_r^2 - \cos\theta \partial_r \left(\frac{\sin\theta}{r} \partial_\theta \right) - \frac{\sin\theta\cos\theta}{r} \partial_\theta \partial_r + \frac{\sin^2\theta}{r} \partial_r + \frac{\sin^2\theta}{r^2} \partial_\theta^2 + \frac{\sin\theta\cos\theta}{r^2} \partial_\theta \\ &+ \sin^2\theta \partial_r^2 + \sin\theta \partial_r \left(\frac{\cos\theta}{r} \partial_\theta \right) + \frac{\sin\theta\cos\theta}{r} \partial_\theta \partial_r + \frac{\cos^2\theta}{r} \partial_r + \frac{\cos^2\theta}{r^2} \partial_\theta^2 - \frac{\sin\theta\cos\theta}{r^2} \partial_\theta \\ &= \partial_r^2 + \frac{1}{r} \partial_r + \frac{\partial_\theta^2}{r^2}, \end{aligned} \tag{5}$$

or equivalently,

$$p^2 = p_r^2 - \frac{\hbar^2}{4r^2} + \frac{L_z^2}{r^2}, \tag{6}$$

where following the Hermitian definition

$$\begin{aligned} p_r &= \frac{1}{2}(\hat{r} \cdot \boldsymbol{p} + \boldsymbol{p} \cdot \hat{r}) = \frac{1}{r} r_i p_i - i\hbar \left(\frac{\partial}{\partial r_i} \frac{r_i}{r} \right) \\ &= -i\hbar \left[\cos\theta \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta} \right) + \sin\theta \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r} \frac{\partial}{\partial \theta} \right) + \frac{1}{2r} \right] \\ &= -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{2r} \right). \end{aligned} \tag{7}$$

Essentially notice that there is no $p^2 = p_r^2 + L^2/r^2$ for polar coordinates in 2 dimensions[1], and as we have shown, it must be modified by an additional factor. Due to the explicit form, it is easy to see that

$$[p^2, L_z] = 0. \tag{8}$$

On the other hand, we need to figure out what $\boldsymbol{p}$ becomes. Focusing on the coordinate transformation, we have corresponding basis transformation as

$$\begin{cases} \hat{r} = \cos\theta\hat{x} + \sin\theta\hat{y} \\ \hat{\theta} = -\sin\theta\hat{x} + \cos\theta\hat{y} \end{cases}, \quad \begin{cases} \hat{x} = \cos\theta\hat{r} - \sin\theta\hat{\theta} \\ \hat{y} = \sin\theta\hat{r} + \cos\theta\hat{\theta} \end{cases}, \tag{9}$$

which is basically a 2D rotation by θ. Then

$$\begin{aligned} \hat{r}_i\partial_i &= (\cos\theta\hat{r} - \sin\theta\hat{\theta})\left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right) + (\sin\theta\hat{r} + \cos\theta\hat{\theta})\left(\sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta}\right) \\ &= \cos^2\theta\hat{r}\partial_r - \frac{\cos\theta\sin\theta}{r}\hat{r}\partial_\theta - \cos\theta\sin\theta\hat{\theta}\partial_r + \frac{\sin^2\theta}{r}\hat{\theta}\partial_\theta \\ &+ \sin^2\theta\hat{r}\partial_r + \frac{\sin\theta\cos\theta}{r}\hat{r}\partial_\theta + \sin\theta\cos\theta\hat{\theta}\partial_r + \frac{\cos^2\theta}{r}\hat{\theta}\partial_\theta \\ &= \hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta, \end{aligned} \tag{10}$$

and equivalently

$$\begin{aligned} \partial_i\hat{r}_i &= \left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)(\cos\theta\hat{r} - \sin\theta\hat{\theta}) + \left(\sin\theta\frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial\theta}\right)(\sin\theta\hat{r} + \cos\theta\hat{\theta}) \\ &= \hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta - \frac{\sin\theta}{r}\frac{\partial\cos\theta\hat{r} - \sin\theta\hat{\theta}}{\partial\theta} + \frac{\cos\theta}{r}\frac{\partial\sin\theta\hat{r} + \cos\theta\hat{\theta}}{\partial\theta} \\ &= \hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta - \frac{\sin\theta}{r}(-\sin\theta\hat{r} + \cos\theta\hat{\theta} - \cos\theta\hat{\theta} + \sin\theta\hat{r}) \\ &+ \frac{\cos\theta}{r}(\cos\theta\hat{r} + \sin\theta\hat{\theta} - \sin\theta\hat{\theta} - \cos\theta\hat{r}) \\ &= \hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta, \end{aligned} \tag{11}$$

where

$$\frac{\partial\hat{r}}{\partial\theta} = -\sin\theta\hat{x} + \cos\theta\hat{y} = \hat{\theta}, \tag{12a}$$

$$\frac{\partial\hat{\theta}}{\partial\theta} = -\cos\theta\hat{x} - \sin\theta\hat{y} = -\hat{r}, \tag{12b}$$

$$\frac{\partial\hat{r}}{\partial r} = 0 = \frac{\partial\hat{\theta}}{\partial\theta} \tag{12c}$$

have been utilized. In the sense we recognize

$$\boldsymbol{p} = -i\hbar\left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta\right) \tag{13}$$

under polar coordinates, which reproduces correctly

$$\begin{aligned}\frac{p^2}{-\hbar^2} &= \left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta\right)\cdot\left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta\right) \\ &= \partial_r^2 + \frac{\hat{\theta}}{r}\frac{\partial \hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta}{\partial\theta} + \frac{1}{r^2}\partial_\theta^2 \\ &= \partial_r^2 + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_\theta^2.\end{aligned} \tag{14}$$

Now let us assume that the non-relativistic free particle is subjected to a perpendicular magnetic field $\boldsymbol{B} = B\hat{z}$. We choose a simple gauge $\boldsymbol{A} = -\boldsymbol{r}\times\boldsymbol{B}/2$ so that $\nabla\times\boldsymbol{A} = \boldsymbol{B}$ is satisfied. We further assume the field to be rotationally symmetric so that $B = B(r)$, and

$$\boldsymbol{A} = \frac{1}{2}rB(r)\hat{\theta} \equiv A(r)\hat{\theta} \tag{15}$$

is accompanied. The first thing we want to show is the canonical angular momentum conservation for free particle under such a symmetric magnetic field. By minimum coupling, we have $H \propto (\boldsymbol{p} + e\boldsymbol{A})^2$ where an electron with charge $-e$ is assumed, and

$$\begin{aligned}\frac{(\boldsymbol{p}+e\boldsymbol{A})^2}{-\hbar^2} &= \left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta + i\frac{e}{\hbar}A(r)\hat{\theta}\right)\cdot\left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta + i\frac{e}{\hbar}A(r)\hat{\theta}\right) \\ &= \partial_r^2 + \frac{\hat{\theta}}{r}\left[\frac{\partial}{\partial\theta}\left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta + i\frac{e}{\hbar}A(r)\hat{\theta}\right)\right] + \left(\frac{1}{r}\partial_\theta + i\frac{e}{\hbar}A(r)\right)^2 \\ &= \left(\partial_r^2 + \frac{1}{r}\partial_r\right) + \left(\frac{1}{r}\partial_\theta + i\frac{e}{\hbar}A(r)\right)^2,\end{aligned} \tag{16}$$

i.e., if we redefine

$$p_r^2 = \partial_r^2 + \frac{1}{r}\partial_r,\ p_\theta = \frac{1}{r}\partial_\theta, \tag{17}$$

we have

$$(\boldsymbol{p}+e\boldsymbol{A})^2 = p_r^2 + (p_\theta + eA(r))^2, \tag{18}$$

and clearly for $L_z = -i\hbar\partial_\theta$,

$$[(\boldsymbol{p}+e\boldsymbol{A})^2, L_z] = 0 \tag{19}$$

stands. Essentially notice that the redefined p_r^2 does not admit a usual p_r.

At the same time, we may look for another quantity as the physical angular momentum, which is defined as

$$L_z^p = [\boldsymbol{r}\times(\boldsymbol{p}+e\boldsymbol{A})]_z = -i\hbar\partial_\theta + e(\boldsymbol{r}\times\boldsymbol{A})_z = L_z + eA(r)(\boldsymbol{r}\times\hat{\theta})_z = L_z + erA(r). \tag{20}$$

This physical angular momentum is not conserved generally for the non-relativistic free particle since

$$[(\boldsymbol{p}+e\boldsymbol{A})^2, L_z^p] = [p_r^2, erA(r)] = e\left(\partial_r^2 + \frac{1}{r}\partial_r\right)(rA(r)), \tag{21}$$

which gives us a typical Euler type differential equation if admitting commutation and leads to solution, by assuming $rA(r) = r^n$ to solve $n = 0$ only, while the second order nature allows for another solution $\ln r$, so that

$$rA(r) = A_0 + A_1 \ln r \Longrightarrow A(r) = \frac{A_0}{r} + \frac{A_1}{r}\ln r. \tag{22}$$

Essentially notice that the A_0/r term describes the vector field away from a vortex line. Importantly, the physical angular momentum is not conserved for non-relativistic free particle subjected to a uniform magnetic field. Also notice that the solution can be got also by decomposing

$$\left(\partial_r + \frac{1}{r}\right)\partial_r f(r) = 0 \Longrightarrow \partial_r f(r) = \frac{A_1}{r} \Longrightarrow f(r) = A_0 + A_1 \ln r. \tag{23}$$

II. Wormhole State

For our consideration, we care about a practical case where a 3D TI is impaled by a half magnetic flux quantum through its center. And here we want to reflect its physics through the minimal continuum model as a modified Dirac equation[2]

$$H(\boldsymbol{k}) = v\boldsymbol{k}\cdot\boldsymbol{\alpha} + (m - bk^2)\beta, \tag{24}$$

where $\alpha_i = \sigma_i\tau_x$, $\beta = \sigma_0\tau_z$ follow the standard representation. For the magnetic flux, we assume that it crosses along z direction and is described by a uniform magnetic field inside a cylinder with radius R_1, i.e.,

$$\boldsymbol{B} = \begin{cases} \dfrac{\Phi_0}{\pi R_1^2}, & r < R_1 \\ 0, & r > R_1 \end{cases}, \ \boldsymbol{A} = \begin{cases} \dfrac{\Phi_0}{2\pi R_1^2} r\hat{\theta}, & r < R_1 \\ \dfrac{\Phi_0}{2\pi r}\hat{\theta}, & r > R_1 \end{cases}, \ \Phi_0 = \frac{h}{2e}. \tag{25}$$

It becomes clear then that through minimal coupling, B_z remains as a good quantum number in the system so that we can separate the Hamiltonian as $H(\boldsymbol{k}) = H_\perp(B_z) + H_\parallel(\boldsymbol{r}, B_z)$, where

$$H_\perp(B_z) = vB_z\alpha_z, \tag{26a}$$

$$H_{\parallel}(\boldsymbol{r}, B_z) = \frac{v}{\hbar}(\boldsymbol{p} + e\boldsymbol{A}) \cdot \boldsymbol{\alpha} + \left[m_k - \frac{b}{\hbar^2}(\boldsymbol{p} + e\boldsymbol{A})^2 \right] \beta, \tag{26b}$$

where $m_k = m - bB_z^2$, and we have redefined the vectors to live in the x-y 2D plane. To express this under polar coordinates, we notice that

$$\hat{r} \cdot \boldsymbol{\sigma} = \cos\theta\sigma_x + \sin\theta\sigma_y = \begin{pmatrix} & e^{-i\theta} \\ e^{i\theta} & \end{pmatrix} = e^{-i\sigma_z\theta}\sigma_x, \tag{27a}$$

$$\hat{\theta} \cdot \boldsymbol{\sigma} = -\sin\theta\sigma_x + \cos\theta\sigma_y = \begin{pmatrix} & -ie^{-i\theta} \\ ie^{i\theta} & \end{pmatrix} = e^{-i\sigma_z\theta}\sigma_y, \tag{27b}$$

which leads to

$$\begin{aligned} (\boldsymbol{p} + e\boldsymbol{A}) \cdot \boldsymbol{\alpha} = \boldsymbol{\alpha} \cdot (\boldsymbol{p} + e\boldsymbol{A}) &= \boldsymbol{\sigma} \cdot \left[-i\hbar \left(\hat{r}\partial_r + \frac{\hat{\theta}}{r}\partial_\theta \right) + eA(r)\hat{\theta} \right] \tau_x \\ &= e^{-i\sigma_z\theta} \left[-i\hbar\sigma_x\partial_r + \sigma_y \left(-i\hbar\frac{1}{r}\partial_\theta + eA(r) \right) \right] \tau_x. \end{aligned} \tag{28}$$

Then totally,

$$\begin{aligned} H_{\parallel}(\boldsymbol{r}, B_z) = v e^{-i\sigma_z\theta} &\left[-i\alpha_x\partial_r + \alpha_y \left(-i\hbar\frac{1}{r}\partial_\theta + eA(r) \right) \right] \\ &+ \left\{ m_k - b \left[\left(\partial_r^2 + \frac{1}{r}\partial_r \right) + \left(\frac{1}{r}\partial_\theta + i\frac{e}{\hbar}A(r) \right)^2 \right] \right\} \beta. \end{aligned} \tag{29}$$

Now for the relativistic linear part, one possible conserved quantity is the total physical angular momentum conservation, i.e.,

$$[(\boldsymbol{p} + e\boldsymbol{A}) \cdot \boldsymbol{\sigma}, J_z^p] \stackrel{?}{=} 0, \;\; J_z^p = L_z^p + \frac{\hbar}{2}\sigma_z. \tag{30}$$

This is the case when

$$
\begin{aligned}
e^{i\sigma_z\theta}\cdot \mathrm{LHS} &= e^{i\sigma_z\theta}\left[e^{-i\sigma_z\theta}\left[-i\hbar\sigma_x\partial_r+\sigma_y\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right],-i\hbar\partial_\theta+erA(r)+\frac{\hbar}{2}\sigma_z\right]\\
&= e^{i\sigma_z\theta}i\hbar\left[\partial_\theta\left(e^{-i\sigma_z\theta}\right)\right]\left[-i\hbar\sigma_x\partial_r+\sigma_y\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right]+(-i\hbar\sigma_x\partial_r)[erA(r)]\\
&+\frac{\hbar}{2}\left\{-i\hbar\partial_r[\sigma_x,\sigma_z]+\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)[\sigma_y,\sigma_z]\right\}\\
&= \hbar\sigma_z\left[-i\hbar\sigma_x\partial_r+\sigma_y\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right]-i\hbar\sigma_x(eA(r)+er\partial_rA(r))\\
&+\frac{\hbar}{2}\left[-2\hbar\sigma_y\partial_r+2i\sigma_x\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right]\\
&= \hbar\left[\hbar\sigma_y\partial_r-i\sigma_x\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right]-ie\hbar\sigma_x(A(r)+r\partial_rA(r))\\
&+\hbar\left[-\hbar\sigma_y\partial_r+i\sigma_x\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right]\\
&= -ie\hbar\sigma_x(A(r)+r\partial_rA(r))=0,
\end{aligned}
\tag{31}
$$

which leads to the solution that

$$
A(r)=\frac{A_0}{r},
\tag{32}
$$

i.e., vector potential away from magnetic flux line. On the other hand, one may see that the canonical total angular momentum is always conserved under symmetric magnetic field as

$$
[(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\sigma},J_z]\Big|_{\boldsymbol{A}=A(r)\hat{\theta}}=0.
\tag{33}
$$

The results can also be seen by brute force index operation, as

$$
\begin{aligned}
[(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\sigma},J_z^p] &= [p_i+eA_i,L_z+e\epsilon_{3mn}r_mA_n]\sigma_i+\frac{\hbar}{2}(p_i+eA_i)[\sigma_i,\sigma_z]\\
&= \epsilon_{3mn}[p_i,r_m]p_n\sigma_i+e\epsilon_{3mn}r_m[A_i,p_n]\sigma_i+e\epsilon_{3mn}[p_i,r_mA_n]\sigma_i+\frac{\hbar}{2}(p_i+eA_i)2i\epsilon_{i3m}\sigma_m\\
&= -i\hbar\epsilon_{3in}p_n\sigma_i+ie\hbar\epsilon_{3mn}r_m(\partial_nA_i)\sigma_i-ie\hbar\epsilon_{3in}A_n\sigma_i-ie\hbar\epsilon_{3mn}r_m(\partial_iA_n)\sigma_i\\
&-i\hbar\epsilon_{3im}p_i\sigma_m+ie\hbar\epsilon_{i3m}A_i\sigma_m\\
&= ie\hbar\epsilon_{3mn}r_m(\partial_nA_i-\partial_iA_n)\sigma_i=ie\hbar\epsilon_{3mn}r_mF_{ni}\sigma_i.
\end{aligned}
\tag{34}
$$

Assuming symmetric magnetic field so that $\boldsymbol{A} = A(r)\hat{\theta}$, and $A_i = -A(r)\epsilon_{ij3}r_j/r$, then

$$\begin{aligned}
\frac{[(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\sigma}, J_z^p]}{ie\hbar} &= \epsilon_{3mn}r_m(\partial_n A_i - \partial_i A_n)\sigma_i \\
&= -\epsilon_{3mn}r_m\left[\partial_n\left(A(r)\epsilon_{ij3}\frac{r_j}{r}\right) - \partial_i\left(A(r)\epsilon_{nj3}\frac{r_j}{r}\right)\right]\sigma_i \\
&= -\epsilon_{3mn}r_m\left[(\partial_r A(r))\frac{r_n}{r}\epsilon_{ij3}\frac{r_j}{r} + A(r)\epsilon_{ij3}\frac{\delta_{nj}}{r} - A(r)\epsilon_{ij3}\frac{r_j r_n}{r^3}\right. \\
&\quad \left. -\epsilon_{nj3}(\partial_r A(r))\frac{r_i}{r}\frac{r_n}{r} - \epsilon_{nj3}A(r)\frac{\delta_{ij}}{r} + \epsilon_{nj3}A(r)\frac{r_n r_j}{r^3}\right]\sigma_i \\
&= -2A(r)\frac{r_i}{r}\sigma_i - \epsilon_{3mn}\epsilon_{nj3}\frac{r_m r_i r_j}{r^3}[r\partial_r A(r) - A(r)] \\
&= -\frac{\boldsymbol{r}\cdot\boldsymbol{\sigma}}{r}[r\partial_r A(r) + A(r)],
\end{aligned} \tag{35}$$

where in the third line we have utilized $\epsilon_{3mn}r_m r_n = 0$ by symmetry. And one sees that this also leads to a flux vector potential for the physical total angular momentum to be conserved. Similarly,

$$\begin{aligned}
[(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\sigma}, J_z] &= [p_i + eA_i, L_z]\sigma_i + \frac{\hbar}{2}(p_i + eA_i)[\sigma_i, \sigma_z] \\
&= \epsilon_{3mn}[p_i, r_m]p_n\sigma_i + e\epsilon_{3mn}r_m[A_i, p_n]\sigma_i + \frac{\hbar}{2}(p_i + eA_i)2i\epsilon_{i3m}\sigma_m \\
&= -i\hbar\epsilon_{3in}p_n\sigma_i + ie\hbar\epsilon_{3mn}r_m(\partial_n A_i)\sigma_i - i\hbar\epsilon_{3im}p_i\sigma_m + ie\hbar\epsilon_{i3m}A_i\sigma_m \\
&= -ie\hbar\epsilon_{3mn}r_m\partial_n\left(A(r)\epsilon_{ij3}\frac{r_j}{r}\right)\sigma_i - ie\hbar\epsilon_{i3m}A(r)\epsilon_{ij3}\frac{r_j}{r}\sigma_m \\
&= -ie\hbar\epsilon_{3mn}r_m\left[(\partial_r A(r))\frac{r_n}{r}\epsilon_{ij3}\frac{r_j}{r} + A(r)\epsilon_{ij3}\frac{\delta_{nj}}{r} - A(r)\epsilon_{ij3}\frac{r_j r_n}{r^3}\right]\sigma_i \\
&\quad - ie\hbar\epsilon_{i3m}A(r)\epsilon_{ij3}\frac{r_j}{r}\sigma_m \\
&= -ie\hbar\frac{A(r)}{r}\left(\epsilon_{3mj}r_m\epsilon_{ij3}\sigma_i + \epsilon_{i3m}\epsilon_{ij3}r_j\sigma_m\right) \\
&= -ie\hbar\frac{A(r)}{r}(\delta_{mi}r_m\sigma_i - \delta_{mj}r_j\sigma_m) = 0,
\end{aligned} \tag{36}$$

so that the canonical total angular momentum is conserved under symmetric magnetic field.

In short, from calculation above, we see that for $\boldsymbol{A} = A(r)\hat{\theta}$,

$$[(\boldsymbol{p}+e\boldsymbol{A})^2, L_z] = 0, \tag{37a}$$

$$[(\boldsymbol{p}+e\boldsymbol{A})^2, L_z^p] = 0, \text{ iff } A(r) = \frac{A_0}{r} + \frac{A_1}{r}\ln r, \tag{37b}$$

$$[(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\sigma}, J_z] = 0, \tag{37c}$$

$$[(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\sigma}, J_z^p] = 0, \text{ iff } A(r) = \frac{A_0}{r}, \tag{37d}$$

with definition for canonical/physical (total) angular momentum

$$\begin{cases} L_z &= (\boldsymbol{r}\times\boldsymbol{p})_z = -i\hbar\partial_\theta \\ L_z^p &= [\boldsymbol{r}\times(\boldsymbol{p}+e\boldsymbol{A})]_z = -i\hbar\partial_\theta + erA(r) \\ J_z &= L_z + \hbar\sigma_z/2 \\ J_z^p &= L_z^p + \hbar\sigma_z/2 \end{cases}. \tag{38}$$

It is trivial to further deduce that $[(\boldsymbol{p}+e\boldsymbol{A})^2, J_z] = 0$, and in the case, the globally conserved quantity for the modified Dirac equation is the canonical total angular momentum under a general polar symmetric magnetic field along z. And especially when we take some ration of half flux quantums inside R_1 as

$$\boldsymbol{A} = \begin{cases} \dfrac{\Phi_0}{2\pi R_1^2} r\hat{\theta}, & r < R_1 \\ \dfrac{\Phi_0}{2\pi r}\hat{\theta}, & r > R_1 \end{cases}, \ \Phi_0 = \nu\frac{h}{2e}, \tag{39}$$

another conserved quantity emerges as the physical total angular momentum when we take $r > R_1$. Notice that this quantity emerges inside R_1 where a uniform magnetic field forms.

In the circumstance, we claim that when $r > R_1$, the eigen-wavefunction can be written as $\Psi_{m_J}(r,\theta) = e^{iL_z\theta}\Psi_{m_J}(r)$, with m_J the good quantum number of total physical angular momentum. To show this, we need firstly find eigenstates of $J_z^p = -i\hbar\partial_\theta + erA(r) + \hbar\sigma_z/2$, and by considering $\sigma_z\eta_z^s = s\eta_z^s$, its eigenstates can be assumed to share the form $\Phi_{m_J}(r,\theta) = \eta_z^s\phi_m(r,\theta)$, with equation

$$\left(-i\hbar\partial_\theta + er\frac{\Phi_0}{2\pi r} + s\frac{\hbar}{2}\right)\phi_m = m_J\phi_m, \tag{40}$$

which can be solved by the trial function $\phi_m = e^{im\theta}$, $m \in \mathbb{Z}$, with solely angular dependence as

$$m + \frac{\nu+s}{2} = m_J. \tag{41}$$

Since m is an integer for single-valuedness, ν integer for flux quantization, and $s = \pm$, we conclude that when ν is odd, m_J is an integer, and vice versa as even for a half-integer. With this information, we can assume the eigenstates of $H_\parallel(\boldsymbol{r}, B_z)$ to be $\Psi_{m_J}(r,\theta,B_z) = e^{iL_z\theta}\Psi_{m_J}(r,B_z)$, as in some sense an extended state of eigenstate of J_z^p, where the angular momentum is taken directly as

$$L_z = m_J - \frac{\nu}{2} - \frac{1}{2}\sigma_z. \tag{42}$$

This trial state can help us to eliminate the angular dependence in eigen problem and turns the equation into a 1D one. Since

$$\begin{aligned}(\boldsymbol{p}+e\boldsymbol{A})\cdot\boldsymbol{\alpha}e^{iL_z\theta} &= e^{-i\sigma_z\theta}\left[-i\hbar\sigma_x\partial_r+\sigma_y\left(-i\hbar\frac{1}{r}\partial_\theta+eA(r)\right)\right]\tau_x e^{i(m_J-\frac{\nu}{2}-\frac{1}{2}\sigma_z)\theta}\\ &= e^{-i\sigma_z\theta}e^{i(m_J-\frac{\nu}{2}+\frac{1}{2}\sigma_z)\theta}\left[-i\hbar\sigma_x\partial_r+\sigma_y\left(\hbar\frac{m_J-\frac{\nu}{2}-\frac{1}{2}\sigma_z}{r}+eA(r)\right)\right]\tau_x\\ &= e^{i(m_J-\frac{\nu}{2}-\frac{1}{2}\sigma_z)\theta}\hbar\left[-i\alpha_x\partial_r+\alpha_y\left(\frac{m_J-\frac{1}{2}\sigma_z}{r}\right)\right],\end{aligned} \tag{43}$$

and

$$\begin{aligned}(\boldsymbol{p}+e\boldsymbol{A})^2\beta e^{iL_z\theta} &= e^{i(m_J-\frac{\nu}{2}-\frac{1}{2}\sigma_z)\theta}(-\hbar^2)\left[\left(\partial_r^2+\frac{1}{r}\partial_r\right)-\left(\frac{m_J-\frac{\nu}{2}-\frac{1}{2}\sigma_z}{r}+\frac{e}{\hbar}A(r)\right)^2\right]\beta\\ &= e^{i(m_J-\frac{\nu}{2}-\frac{1}{2}\sigma_z)\theta}(-\hbar^2)\left[\left(\partial_r^2+\frac{1}{r}\partial_r\right)-\left(\frac{m_J-\frac{1}{2}\sigma_z}{r}\right)^2\right]\beta,\end{aligned} \tag{44}$$

we are able to do transformation from $H_\parallel(\boldsymbol{r},B_z)\Psi_{m_J}(r,\theta,B_z)=E_{m_J}\Psi_{m_J}(r,\theta,B_z)$ to $\tilde{H}_\parallel(r,B_z)\Psi_{m_J}(r,B_z)=E_{m_J}\Psi_{m_J}(r,B_z)$, where

$$\begin{aligned}\tilde{H}_\parallel(r,B_z) &= e^{-iL_z\theta}H_\parallel(\boldsymbol{r},B_z)e^{iL_z\theta}\\ &= v\left[-i\alpha_x\partial_r+\alpha_y\left(\frac{m_J-\frac{1}{2}\sigma_z}{r}\right)\right]+\left\{m_k+b\left[\left(\partial_r^2+\frac{1}{r}\partial_r\right)-\left(\frac{m_J-\frac{1}{2}\sigma_z}{r}\right)^2\right]\right\}\beta\\ &= v\left[-i\alpha_x\left(\partial_r+\frac{1}{2r}\right)+\alpha_y\frac{m_J}{r}\right]+\left\{m_k+b\left[\left(\partial_r^2+\frac{1}{r}\partial_r\right)-\left(\frac{m_J^2+\frac{1}{4}}{r^2}\right)\right]\right\}\beta+b\frac{m_J}{r}\sigma_z\tau_z.\end{aligned} \tag{45}$$

The Hamiltonian can be further block-diagonalized using $\pm\pi/2$ unitary rotations generated by $\sigma_z\tau_y$ and τ_y in succession, i.e.,

$$\beta \xrightarrow{\sigma_z\tau_y} \sigma_z\tau_x,\ \sigma_z\tau_z \xrightarrow{\sigma_z\tau_y} \sigma_0\tau_x, \tag{46a}$$

$$\alpha_x \xrightarrow{\tau_y} \sigma_x\tau_z,\ \alpha_y \xrightarrow{\tau_y} \sigma_y\tau_z,\ \sigma_z\tau_x \xrightarrow{\tau_y} \sigma_z\tau_z,\ \sigma_0\tau_x \xrightarrow{\tau_y} \sigma_z\tau_z, \tag{46b}$$

which makes τ_z a good quantum number for otherwise two identical blocks,

$$h_\parallel(r,B_z)=v\left[-i\sigma_x\left(\partial_r+\frac{1}{2r}\right)+\sigma_y\frac{m_J}{r}\right]+\left\{m_k+b\left[\left(\partial_r^2+\frac{1}{r}\partial_r\right)-\left(\frac{m_J^2+\frac{1}{4}}{r^2}\right)\right]\right\}\sigma_z+b\frac{m_J}{r}\sigma_0. \tag{47}$$

Also notice that under the two consecutive rotations, the z-Hamiltonian becomes

$$H_\perp^R(B_z)=vB_z\sigma_0\tau_x. \tag{48}$$

Now we would like to solve Eq, (47) to give radical 1D solutions, which matter for their energies and serve as basis. Denote its solution as $\psi_{m_J}(r, B_z)$, we have

$$\begin{aligned} h_\parallel \psi_{m_J} &= \begin{pmatrix} m_k + b\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{m_J^2 - m_J + \frac{1}{4}}{r^2}\right) & -i\left(\partial_r + \frac{1}{2r} + \frac{m_J}{r}\right) \\ -i\left(\partial_r + \frac{1}{2r} - \frac{m_J}{r}\right) & -m_k - b\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{m_J^2 + m_J + \frac{1}{4}}{r^2}\right) \end{pmatrix} \begin{pmatrix} \psi_{m_J}^1 \\ \psi_{m_J}^2 \end{pmatrix} \\ &= \begin{pmatrix} m_k + b\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{l^2}{r^2}\right) & -i\left(\partial_r + \frac{l+1}{r}\right) \\ -i\left(\partial_r - \frac{l}{r}\right) & -m_k - b\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{(l+1)^2}{r^2}\right) \end{pmatrix} \begin{pmatrix} f_l \\ g_l \end{pmatrix} \\ &= E_l \begin{pmatrix} f_l \\ g_l \end{pmatrix},\ l = m_J - \frac{1}{2} \in \mathbb{Z} + \frac{1}{2}. \end{aligned} \tag{49}$$

We take trial wave function as

$$\begin{pmatrix} f_l(r) \\ g_l(r) \end{pmatrix} = \begin{pmatrix} c_l B_l(\lambda r) \\ d_l B_{l+1}(\lambda r) \end{pmatrix}, \tag{50}$$

with B_l the Bessel function satisfying the modified Bessel equation of the lth order as

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}x} - \left(1 + \frac{l^2}{x^2}\right) y = 0. \tag{51}$$

Let $x = \lambda r,\ y(x) = B_l(\lambda r)$, and we get

$$\frac{1}{\lambda^2}\left(\frac{\mathrm{d}^2 B_l}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}B_l}{\mathrm{d}r}\right) - \left(1 + \frac{l^2}{\lambda^2 r^2}\right) B_l = 0, \tag{52}$$

i.e.,

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r} - \frac{l^2}{r^2}\right) B_l = \lambda^2 B_l. \tag{53}$$

Now if $R_2 \to +\infty$, we can choose the modified Bessel function to the second kind to count for the vanishment of wavefunction at $r \to +\infty$. Then counting on their iteration relations

$$lK_l + x\frac{\mathrm{d}}{\mathrm{d}x}K_l = -xK_{l-1},\ -lK_l + x\frac{\mathrm{d}}{\mathrm{d}x}K_l = -xB_{l+1}, \tag{54}$$

which are equivalent to

$$\left(\partial_r + \frac{l}{r}\right) K_l = -\lambda K_{l-1},\ \left(\partial_r - \frac{l}{r}\right) K_l = -\lambda K_{l+1}, \tag{55}$$

we are led to

$$
\begin{aligned}
0 &= \begin{pmatrix} m_k - E_l + b\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{l^2}{r^2}\right) & -i\left(\partial_r + \frac{l+1}{r}\right) \\ -i\left(\partial_r - \frac{l}{r}\right) & -(m_k + E_l) - b\left(\partial_r^2 + \frac{1}{r}\partial_r - \frac{(l+1)^2}{r^2}\right) \end{pmatrix} \begin{pmatrix} c_l K_l(\lambda r) \\ d_l K_{l+1}(\lambda r) \end{pmatrix} \\
&= \begin{pmatrix} \left[(m_k - E_l + b\lambda^2)\, c_l + i\lambda d_l\right] K_l \\ \left[i\lambda c_l - (m_k + E_l + b\lambda^2)\right] K_{l+1} \end{pmatrix},
\end{aligned}
\tag{56}
$$

i.e.,

$$
\begin{pmatrix} -E_l + (m_k + b\lambda^2) & i\lambda \\ i\lambda & -E_l - (m_k + b\lambda^2) \end{pmatrix} \begin{pmatrix} c_l \\ d_l \end{pmatrix} = 0, \tag{57}
$$

which leads to

$$
E_l = \pm\sqrt{(m_k + b\lambda^2)^2 - \lambda^2}, \tag{58}
$$

or,

$$
\lambda_{\alpha p} = p\frac{1 - 2m_k b + (-1)^\alpha \sqrt{(2m_k b - 1)^2 - 4b^2(m_k^2 - E^2)}}{2b^2},\ \alpha = 1, 2,\ p = \pm. \tag{59}
$$

The corresponding wavefunction reads

$$
\phi_{l\alpha} = \begin{pmatrix} i\lambda_\alpha K_l(\lambda_\alpha r) \\ [E_l - (m_k + b\lambda_\alpha^2)] K_{l+1}(\lambda_\alpha r) \end{pmatrix}, \tag{60}
$$

where notice that by analytic continuation relation, we abandon the $p = -$ solution since it is not an independent one. Now the vanishing wavefunction boundary condition for $\psi_l = \sum_\alpha C_\alpha^l \phi_{l\alpha}$ at $r = R_1$ leads to

$$
\frac{\lambda_1}{E_l - (m_k + b\lambda_1^2)} \frac{K_l(\lambda_1 R_1)}{K_{l+1}(\lambda_1 R_1)} = \frac{\lambda_2}{E_l - (m_k + b\lambda_2^2)} \frac{K_l(\lambda_2 R_1)}{K_{l+1}(\lambda_2 R_1)}, \tag{61}
$$

and gives

$$
\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} \lambda_2 K_l(\lambda_2 R_1) \\ -\lambda_1 K_l(\lambda_1 R_1) \end{pmatrix}, \tag{62}
$$

which leads to

$$
\psi_l(r) \propto \begin{pmatrix} i\lambda_1\lambda_2 \left[\dfrac{K_l(\lambda_1 r)}{K_l(\lambda_1 R_1)} - \dfrac{K_l(\lambda_2 r)}{K_l(\lambda_2 R_1)}\right] \\ [E_l - (m_k + b\lambda_2^2)] \dfrac{K_{l+1}(\lambda_2 R_1)}{K_l(\lambda_2 R_1)} \left[\dfrac{K_{l+1}(\lambda_1 r)}{K_{l+1}(\lambda_1 R_1)} - \dfrac{K_{l+1}(\lambda_2 r)}{K_{l+1}(\lambda_2 R_1)}\right] \end{pmatrix}. \tag{63}
$$

For the large R behavior, we utilize the large argument asymptotic expansions as

$$
K_l(x) \sim \sqrt{\frac{\pi}{2x}} e^{-x} \left[1 + \sum_{p=1}^{+\infty} \frac{(l, p)}{(2x)^p}\right], \tag{64}
$$

where

$$(l,p) \equiv \frac{\Gamma(1/2+l+p)}{p!\Gamma(1/2+l-p)}. \tag{65}$$

Especially for l integer and half-integer, we have

$$K_n(x) \sim \sqrt{\frac{\pi}{2x}} e^{-x} \sum_{p=0}^{+\infty} \frac{(4n^2-1)(4n^2-9)\cdots(4n^2-(2p-1)^2)}{p!(8x)^p}, \tag{66a}$$

$$K_{n+\frac{1}{2}}(x) \sim \sqrt{\frac{\pi}{2x}} e^{-x} \sum_{p=0}^{n} \frac{(n+p)!}{p!(n-p)!(2x)^p}. \tag{66b}$$

The large x behavior reads correspondingly as

$$\frac{K_{l+1}(x)}{K_l(x)} \sim \begin{cases} 1+\dfrac{2l+1}{2x}, & l \in \mathbb{Z} \\ 1+\dfrac{3l}{4x}, & l \in \mathbb{Z}+\dfrac{1}{2} \end{cases}. \tag{67}$$

For small R behavior, we utilize small x behavior of $K_l(x)$ as

$$K_n(x) \sim \begin{cases} -\ln\left(\dfrac{x}{2}\right) - \gamma, & n \in N^+ \\ \dfrac{(n-1)!}{2}\left(\dfrac{2}{x}\right)^n, & n > 0 \end{cases}, \tag{68}$$

for integer n with γ the Euler-Mascheroni constant, and

$$K_{n+\frac{1}{2}}(x) \sim \sqrt{\frac{\pi}{2x}} e^{-x}. \tag{69}$$

Correspondingly, we have

$$\frac{K_{n+1}(x)}{K_n(x)} \sim \begin{cases} -\dfrac{1}{x\ln(x/2)}, & n = 0 \\ \dfrac{2n}{x}, & n \in N^+ \end{cases}, \tag{70}$$

and

$$\frac{K_{n+1/2}(x)}{K_{n-1/2}(x)} \sim 1. \tag{71}$$

For our purpose to look for a zero mode as $E_l = 0$, we need

$$\frac{K_l(\lambda_1 R_1)}{K_{l+1}(\lambda_1 R_1)} = \frac{K_l(\lambda_2 R_1)}{K_{l+1}(\lambda_2 R_1)}, \tag{72}$$

Considering boundary conditions at R_1 and R_2 leads to the need of modified Bessel functions of the first kind.

III. Fu-Kane Proposal

Fu and Kane proposed p-wave topological superconductivity based on TI surface state in proximity to a superconductor. Their model reads

$$H_{\mathrm{BdG}} = \tau_z(\lambda \boldsymbol{p}\cdot\boldsymbol{\sigma} - \mu) + \Delta\tau_x. \tag{73}$$

Below we will try to solve for the zero mode of the BdG Hamiltonian in the presence of a half-flux vortex.

A. Vortex at origin point — Anti-PBC introduced

In this picture, we assume that the system is infinite, with a half-flux vortex line lying at the very center point of the system with no spreading. The picture is more close to the wormhole effect with a cylinder hollow centering at the vortex line.

In the presence of a half-flux vortex, at $\mu = 0$ and in search of zero mode, we have

$$\left[-i\lambda\tau_z \begin{pmatrix} & e^{-i\theta}\left(\partial_r - \frac{i}{r}\partial_\theta\right) \\ e^{i\theta}\left(\partial_r + \frac{i}{r}\partial_\theta\right) & \end{pmatrix} + \Delta_0(r)\tau_x\right]\phi = 0, \tag{74}$$

where the gauge field does not directly appear, at the cost of introducing anti-periodic boundary condition in θ: $\phi(r,\theta+2\pi) = -\phi(r,\theta)$. Thus, by assuming

$$\phi = \begin{pmatrix} f(r)e^{i\nu\theta} \\ g(r)e^{i(\nu+1)\theta} \end{pmatrix},\ \nu \in \mathbb{Z} + \frac{1}{2}, \tag{75}$$

we obtain

$$-i\lambda\tau_z \begin{pmatrix} e^{i\nu\theta}\left(\partial_r + \frac{\nu+1}{r}\right)g(r) \\ e^{i(\nu+1)\theta}\left(\partial_r - \frac{\nu}{r}\right)f(r) \end{pmatrix} + \Delta_0(r)\tau_x \begin{pmatrix} f(r)e^{i\nu\theta} \\ g(r)e^{i(\nu+1)\theta} \end{pmatrix} = 0. \tag{76}$$

Thus, by taking $\nu = -1/2$ with respect to chiral symmetry, and multiplying τ_x to the left, we have

$$\begin{cases} \Delta_0(r)f(r) - \lambda\tau_y\left(\partial_r + \frac{1}{2r}\right)g(r) = 0 \\ \Delta_0(r)g(r) - \lambda\tau_y\left(\partial_r + \frac{1}{2r}\right)f(r) = 0 \end{cases}. \tag{77}$$

This enlightens us to choose, for instance $g(r) = u(r)\begin{pmatrix}1\\0\end{pmatrix}$, $f(r) = u(r)\begin{pmatrix}0\\-i\end{pmatrix}$, s.t.

$$\left[\lambda\left(\partial_r + \frac{1}{2r}\right) + \Delta_0(r)\right]u(r) = 0. \tag{78}$$

To solve this, we may assume that $u(r) = v(r)e^{-\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda}$, which cancels the $\Delta_0(r)$ term, and leads to an equation about $v(r)$ as

$$v'(r) + \frac{1}{2r}v(r) = 0, \tag{79}$$

which has solution $v(r) = C/r^{1/2}$. Thus, the zero mode reads

$$\phi_\pm(r,\theta) = \frac{C}{r^{1/2}}e^{-\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda}\chi_\pm(\theta), \tag{80}$$

where

$$\chi_+ = \begin{pmatrix} \begin{pmatrix} 0 \\ -i \end{pmatrix} e^{-i\theta/2} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{i\theta/2} \end{pmatrix}, \ \chi_- = \begin{pmatrix} \begin{pmatrix} i \\ 0 \end{pmatrix} e^{-i\theta/2} \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i\theta/2} \end{pmatrix}. \tag{81}$$

The wavefunction is normalizable by

$$\int \mathrm{d}^2x|\phi|^2 = 4\pi|C|^2\int_0^{+\infty} r\mathrm{d}r\frac{1}{r}e^{-2\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda} = 4\pi|C|^2\int_0^{+\infty}\mathrm{d}re^{-2\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda} \approx \frac{2\pi\lambda}{\Delta_0}|C|^2,$$

where we assume a uniform pairing for simplicity.

When the chemical potential is non-zero, we may consider similarly,

$$-i\lambda\tau_z\begin{pmatrix} e^{i\nu\theta}\left(\partial_r + \frac{\nu+1}{r}\right)g(r) \\ e^{i(\nu+1)\theta}\left(\partial_r - \frac{\nu}{r}\right)f(r) \end{pmatrix} + (\Delta_0(r)\tau_x - \mu\tau_z)\begin{pmatrix} f(r)e^{i\nu\theta} \\ g(r)e^{i(\nu+1)\theta} \end{pmatrix} = 0, \tag{82}$$

which gives, after multiplying $i\tau_z$ on the left,

$$\begin{cases} \lambda(\partial_r + \frac{\nu+1}{r})g(r) - (\Delta_0(t)\tau_y + i\mu)f(r) = 0 \\ \lambda(\partial_r - \frac{\nu}{r})f(r) - (\Delta_0(t)\tau_y + i\mu)g(r) = 0 \end{cases}. \tag{83}$$

Then, by letting

$$\begin{cases} f(r) = u(r)e^{-\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda} \\ g(r) = -\tau_y u(r)e^{-\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda} \end{cases}, \tag{84}$$

we eliminate the pairing potential dependence and obtain

$$\begin{cases} \lambda\tau_y(\partial_r + \frac{\nu+1}{r})u(r) + i\mu u(r) = 0 \\ \lambda(\partial_r - \frac{\nu}{r})u(r) + i\mu\tau_y u(r) = 0 \end{cases}. \tag{85}$$

The equation contains a consistent solution when $\nu = -1/2$, which gives a single equation

$$\lambda\tau_y\left(\partial_r + \frac{1}{2r}\right)u(r) + i\mu u(r) = 0. \tag{86}$$

Now we can assume that $u(r) = \begin{pmatrix} u_\uparrow(r) \\ u_\downarrow(r) \end{pmatrix}$, and we have

$$\begin{cases} \lambda(\partial_r + \frac{1}{2r})u_\downarrow(r) + \mu u_\uparrow(r) = 0 \\ \lambda(\partial_r + \frac{1}{2r})u_\uparrow(r) - \mu u_\downarrow(r) = 0 \end{cases}. \tag{87}$$

From the second equation we have $u_\downarrow = \frac{\lambda}{\mu}(\partial_r + \frac{1}{2r})u_\uparrow$, and substituting it into the first equation gives

$$\left(r^2\partial_r^2 + r\partial_r - \frac{1}{4}\right)u_\uparrow + k_F^2 r^2 u_\uparrow = 0, \tag{88}$$

where $k_F = \mu/\lambda$, and by performing substitution $x = k_F r$, we obtain the Bessel equation of the first kind with half-integer order, i.e.,

$$\left[x^2\partial_x^2 + x\partial_x + \left(x^2 - \left(\frac{1}{2}\right)^2\right)\right]u_\uparrow(x) = 0, \tag{89}$$

whose solution reads

$$u_\uparrow(r) = u_\uparrow(x) = J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}}\sin x = \sqrt{\frac{2}{\pi k_F r}}\sin(k_F r). \tag{90}$$

Correspondingly, we have

$$u_\downarrow(r) = \frac{1}{k_F}\left(\partial_r + \frac{1}{2r}\right)u_\uparrow(r) = \sqrt{\frac{2}{\pi k_F r}}\cos(k_F r) \tag{91}$$

Finally, we get the single zero mode solution as

$$\phi(r) = \frac{C}{r^{-1/2}}e^{-\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda}\begin{pmatrix} \begin{pmatrix}\sin(k_F r)\\ \cos(k_F r)\end{pmatrix}e^{-i\theta/2} \\ -\tau_y\begin{pmatrix}\sin(k_F r)\\ \cos(k_F r)\end{pmatrix}e^{i\theta/2}\end{pmatrix}, \tag{92}$$

which holds only when the system is infinite. This solution degrades into χ_+ as $k_F \to 0$. On the other hand, we can choose $u_\uparrow(r) = J_{-1/2}(k_F r)$, and $u_\downarrow(r) = -J_{1/2}(k_F r)$, which gives

$$\phi(r) = \frac{C}{r^{-1/2}}e^{-\int_0^r \mathrm{d}r'\Delta_0(r')/\lambda}\begin{pmatrix} \begin{pmatrix}\cos(k_F r)\\ -\sin(k_F r)\end{pmatrix}e^{-i\theta/2} \\ -\tau_y\begin{pmatrix}\cos(k_F r)\\ -\sin(k_F r)\end{pmatrix}e^{i\theta/2}\end{pmatrix}, \tag{93}$$

which degrades into χ_- as $k_F \to 0$.

B. Spreading vortex with Ginzburg-Landau pairing

In the above picture, a non-spreading vortex introduces two wormhole states — TI zero surface states.

Now we turn to the possibly more realistic case, where the vortex is assumed to spread. We shall also assume that the pairing potential is affected in the Ginzburg-Landau way[3, 4], i.e.,

$$\Delta \to \Delta(r)e^{\pm i\theta}, \tag{94}$$

where possibly for instance $\Delta(r) = \Delta_0 \tanh(r/\xi)$, in the presence of a half-flux vortex (the sign is chosen according to the direction of the magnetic field, and we assume $-$). We then plug this into the BdG Hamiltonian, and at the same time do not consider the gauge field on the normal part. The possible logic or philosophy behind can possibly be that the superconductivity is a Fermi surface property (at least for trivial s-wave pairing), and in the belief that the vortex traps a Majorana zero mode, which is a superconducting state, we only consider the gauge potential influence on the superconducting part, which is transformed onto pairing function in Ginzburg-Landau theory by Gorkov' approach, and not on the normal part. The above 'belief' also comes from the thought that this vortex mode is a topological defect mode, and the defect comes from the pairing potential winding, not the gauge field. To argue this, we may consider $r \to 0^+$, where the gauge field is negligible, while the defect nature indicated by the winding number is still there.

The zero mode solution reads

$$\left[-i\lambda\tau_z e^{-i\sigma_z\theta}(\partial_r\sigma_x + \frac{1}{r}\partial_\theta\sigma_y) + (\Delta(r)\cos\theta\tau_x + \Delta(r)\sin\theta\tau_y - \mu\tau_z)\right]\phi = 0, \tag{95}$$

where ϕ takes PBC in θ direction. Thus, by assuming

$$\phi_l(r,\theta) = e^{i[l-\frac{1}{2}(\tau_z+\sigma_z)\theta]}\phi_l(r),\ l \in \mathbb{Z}, \tag{96}$$

we obtain

$$e^{i[l-\frac{1}{2}(\tau_z+\sigma_z)\theta]}\left[-i\lambda\begin{pmatrix} & \partial_r + \frac{l}{r} & & \\ \partial_r + \frac{1-l}{r} & & & \\ & & & -(\partial_r + \frac{l+1}{r}) \\ & & -(\partial_r - \frac{l}{r}) & \end{pmatrix} + (\Delta(r)\tau_x - \mu\tau_z)\right]\phi(r) = 0. \tag{97}$$

Assume $\phi(r) = e^{-\int_0^r \mathrm{d}r' \Delta(r')/\lambda}\psi(r)$, and we obtain

$$\left[-i\Delta(r)\tau_z\sigma_x - i\lambda \begin{pmatrix} & \partial_r + \frac{l}{r} & & \\ \partial_r + \frac{1-l}{r} & & & \\ & & & -(\partial_r + \frac{l+1}{r}) \\ & & -(\partial_r - \frac{l}{r}) & \end{pmatrix} + \Delta(r)\tau_x - \mu\tau_z\right]\psi(r) = 0. \tag{98}$$

We thus require $-i\tau_z\sigma_x\psi(r) = \tau_x\psi(r)$, and by letting $\psi = (u_\uparrow, u_\downarrow, v_\uparrow, v_\downarrow)$, we have

$$\begin{cases} u_\uparrow = iv_\downarrow \\ u_\downarrow = iv_\uparrow \end{cases} \Leftrightarrow v = -i\sigma_x u. \tag{99}$$

Bring this into the equation, we have

$$\begin{aligned} i\lambda \begin{pmatrix} & \partial_r + \frac{l}{r} \\ \partial_r + \frac{1-l}{r} & \end{pmatrix} u + \mu u = 0, \\ i\lambda \begin{pmatrix} & \partial_r + \frac{l+1}{r} \\ \partial_r - \frac{l}{r} & \end{pmatrix} \sigma_x u + \mu\sigma_x u = 0, \end{aligned} \tag{100}$$

i.e.,

$$\begin{aligned} &\begin{cases} i\lambda(\partial_r + \frac{l}{r})u_\downarrow + \mu u_\uparrow = 0 \\ i\lambda(\partial_r + \frac{1-l}{r})u_\uparrow + \mu u_\downarrow = 0 \end{cases}, \\ &\begin{cases} i\lambda(\partial_r + \frac{l+1}{r})u_\uparrow + \mu u_\downarrow = 0 \\ i\lambda(\partial_r - \frac{l}{r})u_\downarrow + \mu u_\uparrow = 0 \end{cases}, \end{aligned} \tag{101}$$

which contains a solution when $l = 0$:

$$\begin{cases} i\lambda\partial_r u_\downarrow + \mu u_\uparrow = 0 \\ i\lambda(\partial_r + \frac{1}{r})u_\uparrow + \mu u_\downarrow = 0 \end{cases}, \tag{102}$$

which leads to

$$\begin{cases} u_\uparrow = -\frac{i}{k_F}\partial_r u_\downarrow \\ [r^2\partial_r^2 + r\partial_r + (rk_F)^2]u_\downarrow = 0 \end{cases}, \tag{103}$$

where $k_F = \mu/\lambda$. The above equation leads to

$$\begin{cases} u_\downarrow = J_0(k_F r) \\ u_\uparrow = -\frac{i}{k_F}\partial_r J_0(k_F r) = -iJ_1(k_F r) \end{cases}, \tag{104}$$

which gives the zero mode solution as

$$\phi(r,\theta) = Ce^{-\int_0^r \mathrm{d}r'\Delta(r')/\lambda} \begin{pmatrix} -iJ_1(k_F r)e^{-i\theta} \\ J_0(k_F r) \\ -iJ_0(k_F r) \\ -J_1(k_F r)e^{i\theta} \end{pmatrix}. \tag{105}$$

Note that the oddness of J_1 rules out the singularity at $r = 0$ caused by $e^{i\theta}$, while the evenness of J_0 ensures the existence of the bound state at $r = 0$. Note that this is a unique solution with no partner, and decays into χ_- of Fu and Kane's as $k_F \to 0$.

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[1] Paz G. On the connection between the radial momentum operator and the Hamiltonian in n dimensions[J]. European Journal of Physics, 2001, 22(4): 337.

[2] SHUN-QING. SHEN. Topological Insulators: Dirac Equation in Condensed Matter[M]. Springer, 2018.

[3] Fu L, Kane C L. Superconducting proximity effect and majorana fermions<? format?> at the surface of a topological insulator[J]. Physical review letters, 2008, 100(9): 096407.

[4] Chiu C K, Machida T, Huang Y, et al. Scalable Majorana vortex modes in iron-based superconductors[J]. Science Advances, 2020, 6(9): eaay0443.

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